EXPERIMENT 5: Simulate a Cloud Scenario Using CloudSim and Run a Custom Scheduling Algorithm

1. Aim

To simulate a cloud computing environment using CloudSim and demonstrate a custom Shortest Cloudlet First (SCF) scheduling order.

2. Procedure

1. Install a compatible Java Development Kit.
2. Download the CloudSim toolkit and extract the package.
3. Open Eclipse or another Java IDE and create a project named CloudSimSimulation.
4. Add the CloudSim library files and required dependencies to the project build path.
5. Initialize the CloudSim simulation environment in the Java program.
6. Create a Datacenter with hosts, processing elements, memory, storage and bandwidth.
7. Create a DatacenterBroker to manage virtual machines and cloudlets.
8. Create virtual machines with suitable MIPS, RAM, bandwidth and storage values.
9. Create multiple cloudlets with different execution lengths.
10. Implement the custom Shortest Cloudlet First logic by sorting cloudlets in ascending order of length.
11. Submit the virtual machines and the sorted cloudlet list to the broker.
12. Start the CloudSim simulation, retrieve the cloudlet execution details and display the results.
13. Stop the simulation and inspect the scheduling output.

3. Program

Core SCF scheduling demonstration – SCFScheduler.java:

```
import java.util.Arrays;

public class SCFScheduler {
    public static void main(String[] args) {
        int[] cloudlets = {4000, 1000, 3000, 2000};

        System.out.println("Before Scheduling:");
        for (int c : cloudlets) {
            System.out.println(c);
        }

        // Sort cloudlets in ascending order of length
        Arrays.sort(cloudlets);

        System.out.println("\nAfter SCF Scheduling:");
        for (int c : cloudlets) {
            System.out.println("Cloudlet Length: " + c);
        }
    }
}
```

Compile and execute:

```
javac SCFScheduler.java
java SCFScheduler
```

4. Output of Program

```
Before Scheduling:
4000
1000
3000
```

```
2000

After SCF Scheduling:
Cloudlet Length: 1000
Cloudlet Length: 2000
Cloudlet Length: 3000
Cloudlet Length: 4000
```

Note: The code above demonstrates the SCF sorting logic; it is not, by itself, a complete CloudSim datacenter simulation. CloudSim APIs vary by version.

5. Result

Thus, the Shortest Cloudlet First scheduling order was demonstrated by sorting cloudlets according to execution length. For a complete CloudSim experiment, this order must be connected to the CloudSim broker and executed within the configured simulation.

EXPERIMENT 5: Simulate a Cloud Scenario Using CloudSim and Run a Custom Scheduling Algorithm

1. Aim

To simulate a cloud computing environment using the CloudSim toolkit and implement a custom Shortest Cloudlet First (SCF) scheduling algorithm to arrange cloudlets according to their execution lengths.

2. Procedure

1. Install a compatible Java Development Kit on the computer and verify the installation using java -version. 2. Download the CloudSim toolkit from its official source and extract the package. 3. Open Eclipse or another Java-supported integrated development environment. 4. Create a new Java project named CloudSimSimulation. 5. Add the CloudSim libraries and required dependencies to the project's build path. 6. Create a Java class to initialize the CloudSim simulation environment. 7. Configure the simulation by specifying the number of users, calendar and trace settings. 8. Create a Datacenter to represent the cloud infrastructure. 9. Define the required hosts and processing elements with suitable CPU, RAM, storage and bandwidth values. 10. Create a DatacenterBroker to manage communication between the user and the cloud datacenter. 11. Create virtual machines with appropriate processing capacity, memory, bandwidth and storage configurations. 12. Create multiple cloudlets representing user-submitted computing tasks. 13. Assign different execution lengths to the cloudlets so that the scheduling algorithm can arrange them. 14. Implement the Shortest Cloudlet First scheduling logic by sorting the cloudlets in ascending order of their execution lengths. 15. Submit the virtual machines and sorted cloudlet list to the broker. 16. Start the CloudSim simulation and allow the configured cloudlets to execute. 17. Retrieve the cloudlet execution details, including their IDs, status and execution information. 18. Display the scheduling order and simulation results in the console. 19. Stop the simulation and verify that the cloudlets have been processed according to the configured scheduling logic.

4. Output of Program

Input Cloudlet Lengths: 4000 1000 3000 2000 Output Before Scheduling: 4000 1000 3000 2000 Output After Shortest Cloudlet First Scheduling: Cloudlet Length: 1000 Cloudlet Length: 2000 Cloudlet Length: 3000 Cloudlet Length: 4000 Explanation: The SCF scheduling algorithm arranges the cloudlets in ascending order of execution length. The cloudlet requiring the least execution time is placed first, followed by the remaining cloudlets. This demonstrates the basic scheduling order. A complete CloudSim experiment should connect the sorted cloudlets to the broker and execute them within the configured datacenter simulation.

5. Result

Thus, a cloud computing scenario was demonstrated using the CloudSim toolkit. The Shortest Cloudlet First scheduling algorithm was implemented to arrange cloudlets based on their execution lengths, and the expected scheduling order was obtained.